

Explanation

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1 Provided Repository structure Explanation

I analyzed the local folder. It contains two ROS 2 workspaces:

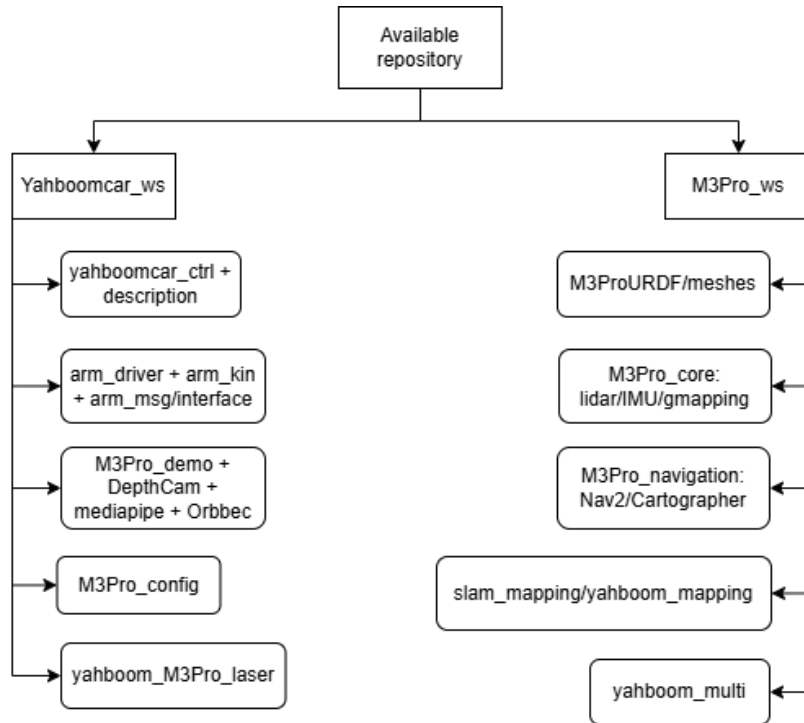


Figure 1: Overview of the downloaded repository structure

1.1 yahboomcar_ws/src

- **yahboomcar_ctrl**: Keyboard and joystick teleoperation. Important files include `yahboom_keyboard.py` and `yahboom_joy_M3Pro.py`.
- **yahboom_M3Pro_description**: Contains the robot URDF, meshes, and RViz model display files.
- **M3Pro_config**: MoveIt2 configuration package containing arm planning settings and fake `ros2_control` configuration.
- **M3Pro_demo**: Includes computer vision and manipulation demonstrations such as AprilTag detection, color tracking, KCF tracking, grasping, and line-following.
- **yahboom_M3Pro_laser**: Lidar-based applications including obstacle avoidance, obstacle warning, and target tracking.
- **yahboom_M3Pro_DepthCam**: Depth camera examples including distance measurement, depth-color visualization, and edge detection.
- **OrbbecSDK_ROS2**: ROS 2 driver package for Orbbec/Dabai depth cameras.
- **arm_interface**, **arm_msgs**: Custom messages and services used for robotic arm communication.
- **arm_kin**: Forward and inverse kinematics services implemented using KDL.
- **arm_driver**: Serial communication driver for the robotic arm using the Rosmaster library.

1.2 M3Pro_ws/src

- **M3Pro**: Robot URDF, meshes, and model display launch files.
- **M3Pro_core**: Core robot packages including laser merging, laser filtering, IMU filtering, and GMapping support.
- **M3Pro_navigation**: Navigation stack configuration including Nav2, Cartographer localization, and RViz navigation setups.
- **slam_mapping**, **yahboom_mapping**: Packages used for SLAM and map generation/saving.
- **ekf_bringup**: Robot localization package that fuses `/odom_raw` and `/imu/data` using an Extended Kalman Filter (EKF).
- **calibration**: Linear and angular calibration utilities based on `/cmd_vel` and TF data.

- **yahboom_multi**: Multi-robot navigation and coordination configurations.
- **patrol**: Laser-based autonomous patrol behavior implementation.
- **largemodel**, **largemodel_arm**: AI, voice interaction, and computer vision application examples.

2 Hardware and software based pipeline

2.1 Hardware pipeline

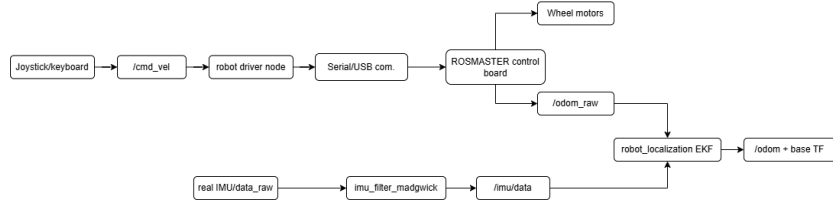


Figure 2: Hardware control pipeline overview

In the hardware-based system, movement commands are generated by a keyboard, joystick, and published on the `/cmd_vel` topic. These commands are sent through the robot driver and serial communication interface to the ROSMASTER control board, which controls the motor driver and wheel motors to move the robot. Wheel encoders provide raw odometry data, while the IMU publishes motion data. The IMU data is processed using the Madgwick filter to obtain a cleaner orientation estimate. Finally, the robotlocalization EKF uses the odometry and IMU data to produce accurate robot pose information (`/odom`) transformation, which are used by RViz2, SLAM, and Nav2 for localization and navigation.

2.2 Software pipeline

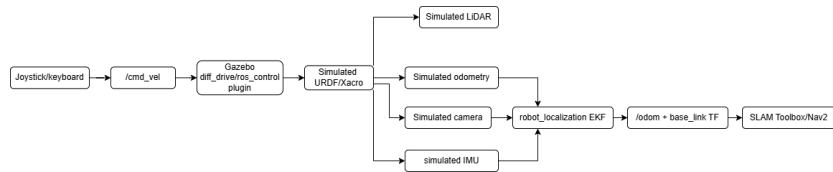


Figure 3: Software control pipeline overview

In the software based system, movement commands are generated by a keyboard and are published on the `/cmd_vel` topic. Instead of being sent to a

physical robot, these commands are processed by a Gazebo `diff_drive` or `ros2_control` plugin, which moves the simulated robot model defined by the URDF/Xacro files. As the robot moves within the virtual environment, Gazebo generates simulated sensor data, including odometry, IMU, LiDAR, and camera information. The odometry and IMU data are used by the `robot_localization` Extended Kalman Filter (EKF) to produce accurate robot pose estimates and the `odom` to `base_link` transformation. The simulated LiDAR data is used by SLAM Toolbox and Nav2 for mapping, localization, and path planning, while the simulated camera data can be utilized for computer vision tasks such as object detection and obstacle avoidance. This simulation-based architecture enables the development, testing, and validation of robotic algorithms without requiring physical hardware.